Recent-Trend Paper

Disclaimer: This is a pattern-based predicted practice paper workflow output, not a leaked, exact, or official CAT paper.

Variant Blueprint

- Variant type: recent_trend
- Strategy: Favor recent and emerging patterns.
- Target section mix: QA 22 from the frozen selected pool; VARC/DILR are outside the current candidate scope
- Expected overlap score: 0.84
- Risk score: 0.35
- Diversity score: 0.727

Questions

Q1. QA - Number System / Divisibility

- Archetype/type: Divisibility/factor-counting problem
- Difficulty: Hard
- Score: 0.808

Find the least positive integer n such that n leaves remainders 3, 5, and 7 when divided by 5, 7, and 9, respectively, and n is divisible by 11.

A. 943 B. 1258 C. 1573 D. 1888

Q2. QA - Geometry / Mensuration

- Archetype/type: Mensuration volume/area problem
- Difficulty: Hard
- Score: 0.811

A right circular cone has base radius 10 cm and height 24 cm. It is cut by a plane parallel to the base so that the curved surface area of the smaller cone removed from the top is one-fourth of the curved surface area of the original cone. The remaining frustum is melted and recast into identical solid cylinders, each of radius 5 cm and height 7 cm. How many such cylinders can be made?

A. 3 B. 4 C. 5 D. 6

Q3. QA - Arithmetic / Profit and Loss

- Archetype/type: Profit-loss transaction problem
- Difficulty: Hard
- Score: 0.91

A shop buys 90 identical bags at Rs 640 each. It marks every bag at the same price. Of the 90 bags, 36 are sold at a 10% discount, 30 at a 20% discount, 12 at a 35% discount because of minor defects, and 12 remain unsold. If the shop still makes an overall profit of Rs 6600 on the entire purchase, what is the marked price of each bag?

A. Rs 900 B. Rs 960 C. Rs 1000 D. Rs 1080

Q4. QA - Arithmetic / Mixtures and Alligation

- Archetype/type: Mixture concentration problem
- Difficulty: Hard
- Score: 0.905

Solution A is 40% acid by volume and Solution B is 70% acid by volume. A chemist mixes a litres of Solution A with b litres of Solution B. After mixing, exactly 10 litres of pure water evaporates from the combined mixture. The remaining mixture has volume 40 litres and is 75% acid. Find the ratio $a : b$.

A. 1 : 1 B. 1 : 2 C. 2 : 3 D. 3 : 5

Q5. QA - Arithmetic / Percentages

- Archetype/type: Successive percentage change
- Difficulty: Hard
- Score: 0.892

A value is first increased by p%, then decreased by 20%, and then again increased by p%. The final value is exactly 90% of what the value would have been after only a single p% increase from the original. If p is positive, find p.

Q6. QA - Arithmetic / Time and Work

- Archetype/type: Work-efficiency problem
- Difficulty: Medium
- Score: 0.882

A can finish a job alone in 30 days, while B can finish it alone in 20 days. A starts the job alone. B joins after 6 days and leaves as soon as only one-eighth of the job is left. A then completes the remaining work alone. In how many days is the job completed?

A. 17.25 days B. 17.85 days C. 18.25 days D. 18.75 days

Q7. QA - Arithmetic / Ratio and Proportion

- Archetype/type: Ratio sharing problem
- Difficulty: Hard
- Score: 0.873

The monthly incomes of P and Q are in the ratio 8:11, while their monthly expenditures are in the ratio 5:7. Q saves Rs 3000 more than P. After P invests

20% of his saving and Q invests 25% of his saving, their remaining savings are equal. What is Q's monthly income in rupees?

Q8. QA - Geometry / Triangles

- Archetype/type: Triangle similarity problem
- Difficulty: Medium-Hard
- Score: 0.865

In triangle ABC, point D lies on side AB and point E lies on side AC such that $DE \parallel BC$. The line through D parallel to AC meets BC at point F. If the area of triangle ADE is 16 and the area of quadrilateral $DECF$ is 40, find the area of triangle DBF.

A. 20 B. 24 C. 25 D. 36

Q9. QA - Arithmetic / Averages

- Archetype/type: Weighted average problem
- Difficulty: Hard
- Score: 0.859

In a batch of 40 candidates, the average score is 69.6. The highest 5 candidates have an average score of 92, and the lowest 7 candidates have an average score of 48. The lowest 7 candidates are replaced by 7 new candidates. If the new average score of all 40 candidates becomes 74.5, find the average score of the 7 new candidates.

Q10. QA - Algebra / Inequalities

- Archetype/type: Inequality optimization problem
- Difficulty: Hard
- Score: 0.855

For integer k with $0 \leq k \leq 15$, how many values of k make both roots of

$$x^2 + (k-3)x + (k-2) = 0$$

distinct negative real numbers?

A. 7 B. 8 C. 9 D. 12

Q11. QA - Algebra / Linear Equations

- Archetype/type: Integer feasibility under hidden divisibility constraint
- Difficulty: Hard
- Score: 0.854

A company has three departments X, Y, and Z. Every employee belongs to exactly one department. The number of employees in X is a positive multiple of 4. The number of employees in Y exceeds that in X by a positive multiple of 3. The number of employees in Z equals exactly half the sum of the employees in X and Y. If the total number of employees in the company is between 100 and 130, inclusive, how many distinct values can the total number of employees take?

Q12. QA - Algebra / Functions

- Archetype/type: Function value/domain problem
- Difficulty: Hard
- Score: 0.852

A function f defined on positive integers satisfies $f(1) = 1$ and

$$f(n) = n - f(f(n-1)) \quad \text{for } n \geq 2.$$

Find $f(20)$.

A. 10 B. 11 C. 12 D. 13

Q13. QA - Algebra / Logarithms

- Archetype/type: Base-change identity with algebraic cancellation
- Difficulty: Hard
- Score: 0.842

Let $a = \log_{12} 18$ and $b = \log_{24} 54$. ; What is the value of

$$\frac{1 - ab}{a - b} \text{ ?}$$

A. 3 B. 5 C. 7 D. -5

Q14. QA - Geometry / Coordinate Geometry

- Archetype/type: Coordinate geometry line/slope/distance problem
- Difficulty: Hard
- Score: 0.839

In a parallelogram $ABCD$ with vertices taken in that order, $A = (1, 2)$ and $B = (7, 5)$. The vertex $C = (p, q)$ has integer coordinates and $p > 10$. The diagonals intersect at a point lying on the line $x + 2y = 16$, and the area of the parallelogram is 42. Find $p + q$.

Q15. QA - Arithmetic / Time Speed Distance

- Archetype/type: Relative speed problem
- Difficulty: Hard
- Score: 0.83

Train A crosses a 150 m platform in 20 seconds. It overtakes a 180 m long train B, moving in the same direction, in 45 seconds. If train A is 250 m long, what is the speed of train B?

A. 32.4 km/h B. 36 km/h C. 37.6 km/h D. 40 km/h

Q16. QA - Algebra / Progressions

- Archetype/type: AP/GP sequence problem
- Difficulty: Hard
- Score: 0.824

An arithmetic progression has positive integer first term a and positive integer common difference d. The sum of the first 7 terms equals the sum of the next 5 terms, i.e. the sum of the 8th to 12th terms. If the 10th term lies strictly between 50 and 80, how many ordered pairs (a, d) are possible?

A. 1 B. 2 C. 3 D. 4

Q17. QA - Number System / Divisibility

- Archetype/type: Divisibility/factor-counting problem
- Difficulty: Hard
- Score: 0.796

How many five-digit numbers of the form $3a7b2$, where a and b are digits, are divisible by 18 but not by 36?

A. 4 B. 5 C. 6 D. 7

Q18. QA - Geometry / Mensuration

- Archetype/type: Mensuration volume/area problem
- Difficulty: Hard
- Score: 0.793

A solid capsule is formed by attaching two hemispheres to the two ends of a right circular cylinder. The cylinder and the hemispheres have the same radius r, and the cylindrical part has height h. If every linear dimension of the capsule is doubled, its surface area increases by 312π and its volume increases by $\frac{3472\pi}{3}$. Find $h - r$.

A. 1 B. 2 C. 3 D. 4

Q19. QA - Number System / Divisibility

- Archetype/type: Divisibility/factor-counting problem
- Difficulty: Hard
- Score: 0.711

How many integers from 1 to 300 are divisible by exactly one of 4, 6 and 10?

A. 60 B. 70 C. 80 D. 90

Q20. QA - Number System / Divisibility

- Archetype/type: Divisibility/factor-counting problem
- Difficulty: Hard
- Score: 0.678

How many integers from 1 to 600 are divisible by exactly one of 6, 10 and 15, but not by all three? The phrase exactly one is to be interpreted literally.

A. 120 B. 130 C. 140 D. 150

Q21. QA - Arithmetic / Profit and Loss

- Archetype/type: Profit-loss transaction problem
- Difficulty: Medium
- Score: 0.908

A retailer buys 100 identical speakers at Rs 600 each and marks all of them at the same price. He sells 50 speakers at a discount of 10%, 30 speakers at a discount of 20%, and the remaining speakers, after minor carton damage, at a discount of 35%. If his overall profit is 9 1/3%, what is the marked price of each speaker?

A. Rs 760 B. Rs 780 C. Rs 800 D. Rs 840

Q22. QA - Arithmetic / Percentages

- Archetype/type: Successive percentage change
- Difficulty: Hard
- Score: 0.887

In January, revenue from product X was exactly 40% of a firm's total revenue. In February, total revenue increased by 25%, but revenue from X decreased by 10% because only some contracts were renewed. By what percentage did revenue from all other products increase from January to February?

Answer Key

Q	Answer	Score
1	(C) 1573	0.808
2	(B) 4	0.811
3	Rs 1000	0.91
4	(B) 1 : 2	0.905
5	12.5	0.892

Q	Answer	Score
6	17.85 days	0.882
7	825000	0.873
8	(C) 25	0.865
9	76	0.859
10	(B) 8	0.855
11	10	0.854
12	(C) 12	0.852
13	(B) 5	0.842
14	23	0.839
15	37.6 km/h	0.83
16	(A) 1	0.824
17	(C) 6	0.796
18	(A) 1	0.793
19	70	0.711
20	140	0.678
21	Rs 800	0.908
22	145/3	0.887

Detailed Solutions

Q1

Step 1: The given remainder conditions are

$$n \equiv 3 \pmod 5, \quad n \equiv 5 \pmod 7, \quad n \equiv 7 \pmod 9.$$

Step 2: Each remainder is 2 less than the corresponding divisor. Hence

$$n + 2$$

is divisible by 5, 7, and 9. Step 3: Since

$$\operatorname{lcm}(5, 7, 9) = 315,$$

we can write

$$n + 2 = 315k,$$

so

$$n = 315k - 2.$$

Step 4: Now impose divisibility by 11:

$$315k - 2 \equiv 0 \pmod{11}.$$

Step 5: Since

$$315 \equiv 7 \pmod{11},$$

we get

$$7k - 2 \equiv 0 \pmod{11},$$

or

$$7k \equiv 2 \pmod{11}.$$

Step 6: The inverse of 7 modulo 11 is 8, because

$$7 \cdot 8 = 56 \equiv 1 \pmod{11}.$$

Therefore,

$$k \equiv 2 \cdot 8 = 16 \equiv 5 \pmod{11}.$$

Step 7: The least positive value is $k = 5$. Hence

$$n = 315(5) - 2 = 1575 - 2 = 1573.$$

Q2

Step 1: The original cone has radius 10 and height 24, so its slant height is

$$\sqrt{10^2 + 24^2} = 26.$$

Step 2: The curved surface area of a cone is proportional to rl. For similar cones, both r and l scale by the same linear factor, so curved surface area scales as the square of the linear factor. Step 3: Since the smaller cone has one-fourth the curved surface area of the original cone, its linear scale factor is

$$\sqrt{\frac{1}{4}} = \frac{1}{2}.$$

Step 4: Therefore the volume of the smaller cone is

$$\left(\frac{1}{2}\right)^3 = \frac{1}{8}$$

of the original cone's volume. Step 5: The original cone's volume is

$$\frac{1}{3}\pi(10)^2(24) = 800\pi.$$

So the frustum volume is

$$800\pi - \frac{1}{8}(800\pi) = 700\pi.$$

Step 6: Each cylinder has volume

$$\pi(5)^2(7) = 175\pi.$$

Therefore the number of cylinders is

$$\frac{700\pi}{175\pi} = 4.$$

Q3

Step 1: Total cost of all 90 bags = 90 x 640 = Rs 57600. Step 2: Profit is calculated on the entire purchase, including the unsold bags, so required revenue = 57600 + 6600 = Rs 64200. Step 3: Let the marked price be M. Only 78 bags are sold, giving revenue 36 x 0.90M + 30 x 0.80M + 12 x 0.65M = 32.4M + 24M + 7.8M = 64.2M. Step 4: Therefore 64.2M = 64200, so M = Rs 1000. Step 5: Verification: revenue = 36 x 900 + 30 x 800 + 12 x 650 = 32400 + 24000 + 7800 = Rs 64200.

Q4

Step 1: After evaporation, the remaining volume is 40 litres and the acid concentration is 75%. Hence acid in the final mixture is

$$0.75 \times 40 = 30 \text{ litres.}$$

Step 2: Evaporation removes only water, not acid. Therefore acid before evaporation was also 30 litres. Step 3: Since 10 litres of water evaporated and final volume is 40 litres, the pre-evaporation mixed volume was

$$40 + 10 = 50 \text{ litres.}$$

So

$$a + b = 50.$$

Step 4: Acid balance before evaporation:

$$0.40a + 0.70b = 30.$$

Step 5: Substitute $a = 50 - b$:

$$0.40(50 - b) + 0.70b = 30,$$

$$20 - 0.40b + 0.70b = 30,$$

$$0.30b = 10 \Rightarrow b = \frac{100}{3}.$$

Then

$$a = 50 - \frac{100}{3} = \frac{50}{3}.$$

Step 6: Therefore

$$a : b = \frac{50}{3} : \frac{100}{3} = 1 : 2.$$

Q5

Step 1: Let the original value be 100. Step 2: After the three-stage change, the value is 100(1+p/100)(0.8)(1+p/100). Step 3: After only a single p% increase, the value would be 100(1+p/100). Step 4: The first value is 90% of the second, so 0.8(1+p/100)^2 = 0.9(1+p/100). Step 5: Since p is positive, divide by the common positive factor to get 0.8(1+p/100)=0.9, so p=12.5.

Q6

A's rate is 1/30 and B's rate is 1/20. In the first 6 days, A completes 6/30 = 1/5 of the job. B joins and they work together at rate 1/30 + 1/20 = 1/12 per day. B leaves when only 1/8 is left, so 7/8 of the job has been completed. The part done together is 7/8 - 1/5 = (35 - 8)/40 = 27/40. Time together = (27/40)/(1/12) = 81/10 = 8.1 days. The remaining 1/8 is done by A alone in (1/8)/(1/30) = 15/4 = 3.75 days. Total time = 6 + 8.1 + 3.75 = 17.85 days.

Q7

Step 1: Let incomes be 8x and 11x, and expenditures be 5y and 7y. Then P's saving is 8x - 5y and Q's saving is 11x - 7y. Step 2: Q saves Rs 3000 more, so 11x - 7y = 8x - 5y + 3000, giving 3x - 2y = 3000. Step 3: After investment, remaining savings are 80% of P's saving and 75% of Q's saving, so 0.8(8x - 5y) = 0.75(11x - 7y). Since Q's saving exceeds P's by 3000, this is also 0.8(SQ - 3000) = 0.75SQ. Hence SQ = 48000. Step 4: Therefore 11x - 7y = 48000 and 3x - 2y = 3000. From the second equation, y = 1.5x - 1500. Step 5: Substitute: 11x - 7(1.5x - 1500) = 48000, so 0.5x + 10500 = 48000, x = 75000. Q's income = 11x = Rs 825000.

Q8

Step 1: Let $\dfrac{AD}{AB} = k$. Since $DE \parallel BC$, triangle $ADE \sim ABC$ with linear ratio k, so $[ADE] = k^2 \cdot [ABC]$. Step 2: Since $DF \parallel AC$, triangle $DBF \sim ABC$ (same angle at B, with $DB/AB = 1-k$ and $BF/BC = 1-k$). Hence $[DBF] = (1-k)^2 \cdot [ABC]$. Step 3: Identify the hidden structure of $DECF$. Both $DE \parallel BC$ and $DF \parallel AC$. By the proportionality theorem, $DE = k \cdot BC$ and $FC = BC - BF = (1-(1-k)) \cdot BC = k \cdot BC$. So $DE = FC$ with $DE \parallel FC$, meaning $DECF$ is a *parallelogram*, not just a trapezoid. Step 4: Since the three regions $[ADE]$, $[DECF]$, $[DBF]$ partition $[ABC]$:

$$[ABC] = [ADE] + [DBF] + [DECF] = k^2T + (1-k)^2T + 2k(1-k)T = T.$$

So $[DECF] = 2k(1-k)\,T$. Step 5: Now $[DECF]^2 = 4k^2(1-k)^2T^2 = 4 \cdot [ADE] \cdot [DBF]$. Substituting:

$$40^2 = 4 \cdot 16 \cdot [DBF] \ \Rightarrow\ [DBF] = \frac{1600}{64} = 25.$$

Q9

Step 1: Convert the original average into the original total:

$$40 \times 69.6 = 2784.$$

Step 2: The total score of the lowest 7 candidates is

$$7 \times 48 = 336.$$

Step 3: After replacing these 7 candidates, the total score of the unchanged 33 candidates is

$$2784 - 336 = 2448.$$

Step 4: The new average of all 40 candidates is 74.5, so the new total score is

$$40 \times 74.5 = 2980.$$

Step 5: Therefore, the total score of the 7 new candidates is

$$2980 - 2448 = 532.$$

Step 6: Hence their average score is

$$\frac{532}{7} = 76.$$

Q10

Step 1: For both roots negative and *distinct* real:

- Discriminant > 0 (strict, for distinct): $(k-3)^2 - 4(k-2) > 0$.
- Sum of roots negative: $-(k-3) < 0$, i.e. $k > 3$.
- Product of roots positive: $k - 2 > 0$, i.e. $k > 2$.

Step 2: Simplify discriminant: $(k-3)^2 - 4(k-2) = k^2 - 6k + 9 - 4k + 8 = k^2 - 10k + 17 > 0$. Step 3: Roots of $k^2 - 10k + 17 = 0$: $k = \frac{10 \pm \sqrt{100-68}}{2} = \frac{10 \pm \sqrt{32}}{2} = 5 \pm 2\sqrt{2}$. Approximately $5 + 2\sqrt{2} \approx 7.83$ and $5 - 2\sqrt{2} \approx 2.17$. Step 4: So $k^2 - 10k + 17 > 0$ iff $k < 5 - 2\sqrt{2}$ or $k > 5 + 2\sqrt{2}$. Step 5: Combine with $k > 3$ and $k > 2$: intersection is $k > 5 + 2\sqrt{2} \approx 7.83$ (the branch $k < 5 - 2\sqrt{2} \approx 2.17$ is killed by $k > 3$). Step 6: Integer k with $7.83 < k \leq 15$: $k \in \{8, 9, 10, 11, 12, 13, 14, 15\}$. Count $= 8$. Step 7: Verify endpoints. $k = 8$: $x^2 + 5x + 6 = (x+2)(x+3) = 0$, roots $-2, -3$, both negative and distinct ✓. $k = 7$: $x^2 + 4x + 5 = 0$, discriminant $= 16 - 20 = -4 < 0$, complex roots $\times$ (good).

Q11

Step 1: Let the number of employees in X be $4a$ where $a \geq 1$, and let Y exceed X by $3b$ where $b \geq 1$. Then $|Y| = 4a + 3b$. Step 2: The number of employees in Z is

$$|Z| = \frac{|X| + |Y|}{2} = \frac{4a + 4a + 3b}{2} = \frac{8a + 3b}{2}.$$

For $|Z|$ to be a positive integer, $8a + 3b$ must be even. Since $8a$ is always even, $3b$ must be even, which forces b to be even. Write $b = 2k$ with $k \geq 1$. Step 3: With $b = 2k$, we get $|Z| = 4a + 3k$ and

$$T = |X| + |Y| + |Z| = 4a + (4a + 6k) + (4a + 3k) = 12a + 9k = 3(4a + 3k).$$

So T is always a multiple of 3. Step 4: Impose $100 \leq T \leq 130$:

$$100 \leq 3(4a+3k) \leq 130 \implies 34 \leq 4a+3k \leq 43.$$

Let $m = 4a+3k$ where $a \geq 1$ and $k \geq 1$, so $m \geq 7$. For any integer $m \geq 7$, a representation $m = 4a+3k$ with $a, k \geq 1$ exists: if $m \equiv 1 \pmod 3$, take $a = 1$; if $m \equiv 2 \pmod 3$, take $a = 2$; if $m \equiv 0 \pmod 3$, take $a = 3$; and solve for k accordingly. Each resulting k is positive for $m \geq 7$. Step 5: The integers m from 34 to 43 inclusive give $43 - 34 + 1 = 10$ values of m, hence 10 distinct totals $T \in \{102, 105, 108, 111, 114, 117, 120, 123, 126, 129\}$.

Q12

Step 1: Compute $f(n)$ step by step. $f(1) = 1.\backslash\ f(2) = 2 - f(f(1)) = 2 - f(1) = 1.\backslash\ f(3) = 3 - f(f(2)) = 3 - f(1) = 2.\backslash\ f(4) = 4 - f(f(3)) = 4 - f(2) = 3.\backslash\ f(5) = 5 - f(f(4)) = 5 - f(3) = 3.\backslash\ f(6) = 6 - f(f(5)) = 6 - f(3) = 4.\backslash\ f(7) = 7 - f(f(6)) = 7 - f(4) = 4.\backslash\ f(8) = 8 - f(f(7)) = 8 - f(4) = 5.$ Step 2: Continue. $f(9) = 9 - f(f(8)) = 9 - f(5) = 6.\backslash\ f(10) = 10 - f(f(9)) = 10 - f(6) = 6.\backslash\ f(11) = 11 - f(f(10)) = 11 - f(6) = 7.\backslash\ f(12) = 12 - f(f(11)) = 12 - f(7) = 8.$ Step 3: Continue. $f(13) = 13 - f(f(12)) = 13 - f(8) = 8.\backslash\ f(14) = 14 - f(f(13)) = 14 - f(8) = 9.\backslash\ f(15) = 15 - f(f(14)) = 15 - f(9) = 9.\backslash\ f(16) = 16 - f(f(15)) = 16 - f(9) = 10.$ Step 4: Continue. $f(17) = 17 - f(f(16)) = 17 - f(10) = 11.\backslash\ f(18) = 18 - f(f(17)) = 18 - f(11) = 11.\backslash\ f(19) = 19 - f(f(18)) = 19 - f(11) = 12.\backslash\ f(20) = 20 - f(f(19)) = 20 - f(12) = 20 - 8 = 12.$ Step 5: $f(20) = 12$.

Q13

Step 1: Set $p = \ln 2$ and $q = \ln 3$. Then

$$a = \frac{\ln 18}{\ln 12} = \frac{p+2q}{2p+q}, \qquad b = \frac{\ln 54}{\ln 24} = \frac{p+3q}{3p+q}.$$

Step 2: Compute $a - b$ over the common denominator $(2p+q)(3p+q)$:

$$a - b = \frac{(p+2q)(3p+q) - (p+3q)(2p+q)}{(2p+q)(3p+q)}.$$

Expanding the numerator:

$$(3p^2 + 7pq + 2q^2) - (2p^2 + 7pq + 3q^2) = p^2 - q^2 = (p-q)(p+q).$$

Step 3: Compute ab:

$$ab = \frac{(p+2q)(p+3q)}{(2p+q)(3p+q)} = \frac{p^2 + 5pq + 6q^2}{6p^2 + 5pq + q^2}.$$

Step 4: Compute $1 - ab$:

$$1 - ab = \frac{(6p^2 + 5pq + q^2) - (p^2 + 5pq + 6q^2)}{(2p+q)(3p+q)} = \frac{5p^2 - 5q^2}{(2p+q)(3p+q)} = \frac{5(p-q)(p+q)}{(2p+q)(3p+q)}.$$

Step 5: Form the ratio:

$$\frac{1-ab}{a-b} = \frac{5(p-q)(p+q)}{(2p+q)(3p+q)} \cdot \frac{(2p+q)(3p+q)}{(p-q)(p+q)} = 5.$$

Note that $p \neq q$ (since $\ln 2 \neq \ln 3$) and $p+q \neq 0$, so the cancellation is valid.

Q14

Step 1: In a parallelogram, the diagonals bisect each other. Hence the midpoint of AC is

$$\left(\frac{1+p}{2}, \frac{2+q}{2}\right).$$

Step 2: This point lies on $x+2y=16$, so

$$\frac{1+p}{2} + 2\left(\frac{2+q}{2}\right) = 16.$$

Step 3: Simplify:

$$\frac{1+p}{2} + 2 + q = 16.$$

Multiplying by 2,

$$1+p+4+2q = 32,$$

so

$$p+2q = 27.$$

Step 4: The adjacent side vectors are

$$\overrightarrow{AB} = (6,3), \qquad \overrightarrow{BC} = (p-7, q-5).$$

The area of the parallelogram is the absolute value of their determinant:

$$|6(q-5) - 3(p-7)| = 42.$$

Step 5: Substitute $p = 27-2q$:

$$|6q - 30 - 3(27-2q-7)| = 42.$$

This becomes

$$|6q-30-3(20-2q)| = 42,$$

or

$$|12q - 90| = 42.$$

Step 6: Therefore,

$$12q - 90 = 42 \quad \text{or} \quad 12q - 90 = -42.$$

So

$$q = 11 \quad \text{or} \quad q = 4.$$

Step 7: From $p + 2q = 27$, these give

$$(q, p) = (11, 5) \quad \text{or} \quad (4, 19).$$

The condition $p > 10$ selects

$$p = 19, \qquad q = 4.$$

Hence

$$p + q = 23.$$

Q15

Step 1: While crossing the platform, Train A covers its own length plus platform length = 250 + 150 = 400 m in 20 seconds. So speed of A = 20 m/s. Step 2: While overtaking Train B in the same direction, Train A must cover the sum of both train lengths relative to B: 250 + 180 = 430 m. Step 3: Relative speed = 430/45 = 86/9 m/s. Step 4: Speed of B = 20 - 86/9 = 94/9 m/s. Step 5: Convert to km/h: (94/9) x 18/5 = 188/5 = 37.6 km/h.

Q16

Step 1: Sum of the first 7 terms is

$$S_7 = \frac{7}{2}(2a + 6d) = 7a + 21d.$$

Step 2: Sum of the 8th to 12th terms is

$$(a + 7d) + (a + 8d) + (a + 9d) + (a + 10d) + (a + 11d) = 5a + 45d.$$

Step 3: Equating the two sums,

$$7a + 21d = 5a + 45d \Rightarrow 2a = 24d \Rightarrow a = 12d.$$

Step 4: The 10th term is

$$a + 9d = 12d + 9d = 21d.$$

The condition $50 < 21d < 80$ gives

$$\frac{50}{21} < d < \frac{80}{21}.$$

Step 5: Since d is a positive integer, the only possible value is $d = 3$. Then $a = 36$. Hence the only ordered pair is $(36, 3)$.

Q17

Step 1: Since the last digit is 2, the number is even. Thus divisibility by 18 reduces to divisibility by 9 along with evenness. Step 2: The digit sum is

$$3 + a + 7 + b + 2 = 12 + a + b.$$

Step 3: For divisibility by 9,

$$12 + a + b$$

must be a multiple of 9. Since $a + b$ ranges from 0 to 18, the possible values are

$$a + b = 6 \quad \text{or} \quad a + b = 15.$$

Step 4: Since $36 = 4 \times 9$, a number already divisible by 9 is divisible by 36 precisely when its last two digits form a multiple of 4. Step 5: The last two digits are $b2$, representing $10b + 2$. Modulo 4,

$$10b + 2 \equiv 2b + 2 \pmod 4.$$

This is divisible by 4 when b is odd. Therefore, not divisible by 36 requires b to be even. Step 6: For $a + b = 6$ with b even, the valid pairs are

$$(b, a) = (0, 6), (2, 4), (4, 2), (6, 0),$$

giving 4 pairs. Step 7: For $a + b = 15$ with b even, the valid pairs are

$$(b, a) = (6, 9), (8, 7),$$

giving 2 pairs. Step 8: Therefore the total number of valid numbers is

$$4 + 2 = 6.$$

Q18

Step 1: The capsule consists of a cylinder and two hemispheres, which together make one sphere. Its surface area is

$$S = 2\pi rh + 4\pi r^2.$$

Step 2: When all linear dimensions are doubled, surface area becomes four times the original. Hence the increase is

$$4S - S = 3S.$$

Given $3S = 312\pi$, we get

$$S = 104\pi.$$

Thus

$$2\pi rh + 4\pi r^2 = 104\pi,$$

or

$$rh + 2r^2 = 52.$$

Step 3: The volume of the capsule is

$$V = \pi r^2 h + \frac{4}{3}\pi r^3.$$

Step 4: When all linear dimensions are doubled, volume becomes eight times the original. Hence the increase is

$$8V - V = 7V.$$

Given

$$7V = \frac{3472\pi}{3},$$

we get

$$V = \frac{496\pi}{3}.$$

Step 5: Therefore,

$$\pi r^2 h + \frac{4}{3}\pi r^3 = \frac{496\pi}{3}.$$

Cancel π:

$$r^2 h + \frac{4}{3}r^3 = \frac{496}{3}.$$

Step 6: From

$$rh + 2r^2 = 52,$$

multiply by r:

$$r^2 h + 2r^3 = 52r.$$

Substitute $r^2 h = 52r - 2r^3$ into the volume equation:

$$52r - 2r^3 + \frac{4}{3}r^3 = \frac{496}{3}.$$

Step 7: Multiply by 3:

$$156r - 6r^3 + 4r^3 = 496,$$

so

$$r^3 - 78r + 248 = 0.$$

Testing the positive integral factor gives

$$r = 4.$$

Step 8: Use

$$rh + 2r^2 = 52.$$

With $r = 4$,

$$4h + 32 = 52,$$

so

$$h = 5.$$

Therefore,

$$h - r = 5 - 4 = 1.$$

Q19

Step 1: Multiples are 75,50,30. Step 2: Pairwise lcm counts are 25,15,10. Step 3: Triple lcm 60 gives 5. Step 4: Exactly one = singles -2(pair overlaps)+3(triple). Step 5: 155-100+15=70.

Q20

Step 1: Counts of multiples of 6,10,15 are 100,60,40. Step 2: Pairwise lcm counts are lcm(6,10)=30 gives 20, lcm(6,15)=30 gives 20, and lcm(10,15)=30 gives 20. Step 3: Triple lcm is 30, count 20. Step 4: Exactly one count is singles -2(pairwise overlaps) +3(triple overlaps). Step 5: So 200 - 120 + 60 = 140.

Q21

Total cost price = 100 x 600 = Rs 60000. Let the common marked price be M. Revenue from the three groups is 50 x 0.90M + 30 x 0.80M + 20 x 0.65M = 45M + 24M + 13M = 82M. Overall profit is 9 1/3% = 28/3%, so required total selling price = 60000 x (1 + 7/75) = 60000 x 82/75 = Rs 65600. Hence 82M = 65600, so M = Rs 800. Verification: revenue = 50 x 720 + 30 x 640 + 20 x 520 = 36000 + 19200 + 10400 = Rs 65600.

Q22

Step 1: Let January total revenue be 100, so X revenue is 40 and other revenue is 60. Step 2: February total revenue is 125 after the 25% increase. Step 3: February X revenue is 90% of 40, which is 36. Step 4: Therefore February revenue from other products is 125 - 36 = 89. Step 5: The increase in other-products revenue is 29 on a base of 60, so the percentage increase is 2900/60 = 145/3%.

Evidence Lineage

Q	Candidate	Source Question IDs
1	CAT_QA_SPEC_20_GPT55Thinking_AVG_DIV_08	CAT_2019_S2_QA_Q06;CAT_2022_S2_QA_Q13
2	CAT_QA_SPEC_19_GPT55Thinking_COORDINATES_05	CAT_2017_S1_QA_Q18;CAT_2017_S1_QA_Q19
3	CAT_QA_SPEC_01_ELITE_03	CAT_2017_S1_QA_Q12;CAT_2017_S2_QA_Q07
4	CAT_QA_SPEC_10_CORRECTED_04	CAT_2018_S1_QA_Q01;CAT_2018_S1_QA_Q14
5	CAT_QA_SPEC_06_ELITE2_03	CAT_2017_S1_QA_Q01;CAT_2017_S1_QA_Q05
6	CAT_QA_SPEC_02_CAND_01	CAT_2017_S1_QA_Q02;CAT_2017_S1_QA_Q15
7	CAT_QA_SPEC_03_ELITE_01	CAT_2017_S1_QA_Q13;CAT_2018_S1_QA_Q26
8	CAT_QA_SPEC_07_CLAUDE_01	CAT_2017_S1_QA_Q16;CAT_2018_S1_QA_Q09
9	CAT_QA_SPEC_15_GPT55Thinking_AVG_DIV_01	CAT_2017_S1_QA_Q02;CAT_2017_S2_QA_Q26
10	CAT_QA_SPEC_08_CLAUDE_07	CAT_2017_S1_QA_Q03;CAT_2017_S1_QA_Q26
11	CAT_QA_SPEC_12_GENSPARK_FINAL_01	CAT_2017_S1_QA_Q28;CAT_2018_S2_QA_Q29
12	CAT_QA_SPEC_13_CLAUDE_04	CAT_2018_S2_QA_Q30;CAT_2019_S1_QA_Q09

Q	Candidate	Source Question IDs
13	CAT_QA_SPEC_16_GENSPARK_FINAL_Q1	CAT_2017_S1_QA_Q23;CAT_2018_S1_QA_Q11
14	CAT_QA_SPEC_18_GPT5_Thinking_COORDMENTS_04	CAT_2017_S1_QA_Q24;CAT_2019_S1_QA_Q03
15	CAT_QA_SPEC_04_ELITE_01	CAT_2017_S1_QA_Q04;CAT_2017_S2_QA_Q04
16	CAT_QA_SPEC_11_CORRECTED_01	CAT_2017_S1_QA_Q15;CAT_2017_S2_QA_Q21
17	CAT_QA_SPEC_20_GPT5_Thinking_AVG_DIV_07	CAT_2019_S2_QA_Q06;CAT_2022_S2_QA_Q13
18	CAT_QA_SPEC_19_GPT5_Thinking_COORDMENTS_08	CAT_2017_S1_QA_Q18;CAT_2017_S1_QA_Q19
19	CAT_QA_SPEC_20_FULL_01	CAT_2019_S2_QA_Q06;CAT_2022_S2_QA_Q13
20	CAT_QA_SPEC_20_ELITE_2_01	CAT_2019_S2_QA_Q06;CAT_2022_S2_QA_Q13
21	CAT_QA_SPEC_01_CAND_01	CAT_2017_S1_QA_Q08;CAT_2017_S1_QA_Q10
22	CAT_QA_SPEC_06_ELITE_2_01	CAT_2017_S1_QA_Q01;CAT_2017_S1_QA_Q05